The Inclusion-Exclusion Principle I

Example

Count the permutations $i_1 i_2 \cdots i_n$ of $\{1, 2, \cdots, n\}$ in which 1 is not in the first position (that is, $i_1 \neq 1$).

1. Observe that the permutation with 1 not in the first position can be divided into $n-1$ parts according to which of the $n-1$ integers k from $\{2, 3, \ldots, n\}$ is in the first position.
2. A permutation with k in the first position consist of k followed by a permutation of the $(n-1)$-element set $\{1, \ldots, k-1, k+1, \ldots, n\}$.
3. There are $(n-1)!$ permutations of $\{1, 2, \ldots, n\}$ with k in the first position
4. By the addition principle there are $(n-1)!(n-1)$ permutations of $\{1, 2, \ldots, n\}$ with 1 not in the first position.

Alternatively,



The Inclusion-Exclusion Principle II

1. We could make an indirect count by observing that the number of permutations of $\{1, 2, \ldots, n\}$ with 1 in the first position is the same as the number $(n-1)!$ of permutations of $\{2, 3, \ldots, n\}$.
2. Since the total number of permutations of $\{1, 2, \ldots, n\}$ is $n!$. The number of permutations of $\{1, 2, \ldots, n\}$ in which 1 is not in the first position is $n! - (n-1)! = (n-1)!(n-1)$.

Example

Count the number of integers between 1 and 600, inclusive, which are not divisible by 6.

Solution

$600 - 100(= 600/6) = 500.$



The Inclusion-Exclusion Principle III

- If A is a subset of a set S, then the number of objects in A equals the number of objects in S minus the number not in A. Recall that $\overline{A} = S - A$ is the complement of A in S — that is, the set consisting of those objects in S which are not in A. The rule can then be written as

$$|A| = |S| - |\overline{A}| \quad \text{or, equivalently,} \quad |\overline{A}| = |S| - |A|.$$

  This formula is the simplest instance of the inclusion-exclusion principle.

- As a first generalization, let S be a finite set of objects, and let P_1 and P_2 be two "properties" that each object in S may or may not possess. We wish to count the number objects in S that have neither property P_1 nor property P_2.
  1. Let A_1 be the subset of objects of S which have property P_1 and let A_2 be the subsets of objects of S which have property P_2.
  2. Then $\overline{A_1}$ consists of those objects of S not having property P_1 and $\overline{A_2}$ consists of those objects of S not having property P_2.
  3. The objects of the set $\overline{A_1} \cap \overline{A_2}$ are those having neither property P_1 nor property P_2.
  4. We can do this by first including all objects of S in our count ($|S|$),



The Inclusion-Exclusion Principle IV

5. then excluding all objects that have property P_1 ($|A_2|$) and excluding all objects which have property P_2 ($|A_2|$),
6. and then, noting that we have excluded objects having both properties P_2 and P_2 twice, readmitting all such objects once ($|A_1 \cap A_2|$).
7. We then have

$$|\overline{A_1} \cap \overline{A_2}| = |S| - |A_1| - |A_2| + |A_1 \cap A_2|.$$

A object x in S.

Property	Contribution to $\lvert S\rvert - \lvert A_1\rvert - \lvert A_2\rvert + \lvert A_1 \cap A_2\rvert$
neither P_1 nor P_2	1 - 0 - 0 + 0 = 1
only P_1	1 - 1 - 0 + 0 = 0
only P_2	1 - 0 - 1 + 0 = 0
P_1 and P_2	1 - 1 - 1 + 1 = 0



The Inclusion-Exclusion Principle V

- More generally, let $P_1, P_2, \ldots, P_m$ be m properties referring to the objects in S and let

$$A_i = \{x : x \text{ in } S \text{ and } x \text{ has property } P_i\}, \qquad (i = 1, 2, \ldots, m)$$

be the subset of objects of S which have property P_i (and possibly other properties).



The Inclusion-Exclusion Principle VI

Theorem

Let number of objects of S that have none of the properties $P_1, P_2, \ldots, P_m$ is given by

$$|\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_m}| = |S| - \sum |A_i| + \sum |A_i \cap A_j| - \sum |A_i \cap A_j \cap A_k| + \cdots + (-1)^m |A_1 \cap A_2 \cap \cdots \cap A_m| \quad (6.1)$$

where the first sum is over all 1-combinations of $\{1, 2, \ldots, m\}$, the second sum is over all 2-combinations $\{i, j\}$ of $\{1, 2, \ldots, m\}$, the third sum is over all 3-combinations $\{i, j, k\}$ of $\{1, 2, \ldots, m\}$, and so on. The number of terms on the right side of (6.1) is

$$\binom{m}{0} + \binom{m}{1} + \binom{m}{2} + \cdots + \binom{m}{m} = 2^m.$$

1. The left side of equation (6.1) counts the number of objects of S with none of the properties.



The Inclusion-Exclusion Principle VII

2. We can establish the validity of the equation by showing that an object with none of the properties $P_1, P_2, \ldots, P_m$ makes a net contribution of 1 to the right side, and an object with at least one of the properties makes a net contribution of 0.
3. First, consider an object x with none of the properties. Its contribution to the right side of (6.1) is

$$1 - 0 + 0 - 0 + \cdots + (-1)^m 0 = 1$$

since it is in S but in none of the other sets.
4. Now consider an object y with exactly $n \geq 1$ of the properties.
5. The contribution of y to $|S|$ is $1 = \binom{n}{0}$.
6. Its contribution to $\sum |A_i|$ is $n = \binom{n}{1}$ since it has exactly n of the properties and so is a member of exactly n of the sets $A_1, A_2, \ldots, A_m$.
7. The contribution of y to $\sum |A_i \cap A_j|$ is $\binom{n}{2}$ since we may set a pair of the properties y has in $\binom{n}{2}$ ways, and so y is a member of exactly $\binom{n}{2}$ of the sets $A_i \cap A_j$.
8. The contribution of y to $\sum |A_i \cap A_j \cap A_k|$ is $\binom{n}{3}$ and so on.



The Inclusion-Exclusion Principle VIII

9. Thus the net contribution of y to the right side of (6.1) is

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \cdots + (-1)^m \binom{n}{m},$$

which equals

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \cdots + (-1)^n \binom{n}{n},$$

since $n \leq m$ and $\binom{n}{k} = 0$ if $k > n$.

10. Since this last expression equals 0 according to the identity (5.4), the net contribution of y to the right side of (6.1) is 0 if y has at least one of the properties, and the theorem is proved. □



The Inclusion-Exclusion Principle IX

Corollary

The number of objects of S which have at least one of the properties $P_1, P_2, \ldots, P_m$ is given by

$$|A_1 \cup A_2 \cup \cdots \cup A_m| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots + (-1)^{m+1} |A_1 \cap A_2 \cap \cdots \cap A_m|, \tag{6.2}$$

where the summation are as specified in the previous theorem

Example

Count the number of integers between 1 and 1,000, inclusive, that are not divisible by 5, 6 and 8.



The Inclusion-Exclusion Principle X

1. For a real number r, we recall that $\lfloor r \rfloor$ stands for the largest integer which does not exceed r. Also, we shall abbreviate the least common multiple of two integers, a, b, or three integers, a, b, c, by $\mathrm{lcm}(a, b)$ and $\mathrm{lcm}(a, b, c)$, respectively.
2. Let P_1, P_2, and P_3 be the properties that an integer is divisible by 5, 6, and 8, respectively.
3. Let S be the set consisting of the first thousand positive integers.
4. For $i = 1, 2, 3$ let A_i be the set consisting of those integers in S with property P_i.
5. We wish to find the number of integers in $\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}$.
6. We first see that

$$|A_1| = \left\lfloor \frac{1,000}{5} \right\rfloor = 200,$$

$$|A_2| = \left\lfloor \frac{1,000}{6} \right\rfloor = 166,$$

$$|A_3| = \left\lfloor \frac{1,000}{8} \right\rfloor = 125.$$



The Inclusion-Exclusion Principle XI

7. An integer is divisible by both 5 and 6 if and only if it is divisible by $\text{lcm}(5,6)$, and so on. Since $\text{lcm}(5,6) = 30$, $\text{lcm}(5,8) = 40$, $\text{lcm}(6,8) = 24$, we see that

$$|A_1 \cap A_2| = \left\lfloor \frac{1,000}{30} \right\rfloor = 33,$$
$$|A_1 \cap A_3| = \left\lfloor \frac{1,000}{40} \right\rfloor = 25,$$
$$|A_2 \cap A_3| = \left\lfloor \frac{1,000}{24} \right\rfloor = 41.$$

8. Because $\text{lcm}(5,6,8) = 120$, we conclude that

$$|A_1 \cap A_2 \cap A_3| = \left\lfloor \frac{1,000}{120} \right\rfloor = 8.$$

9. Thus, by the inclusion-exclusion principle, the number of integers between 1 and 1,000 that are divisible by none of 5, 6, and 8 equals

$$\begin{aligned} |\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| &= 1,000 - (200 + 166 + 125) + (33 + 25 + 41) - 8 \\ &= 600. \end{aligned}$$



The Inclusion-Exclusion Principle XII

Example

How many permutations of the letters

M, A, T, H, I, S, F, U, N

are there such that none of the words MATH, IS, and FUN occur as consecutive letters? (Thus, for instance, the permutation MATH-IS-FUN is not allowed nor are the permutations INUMATHSF and ISMATHFUN.)

1. We identify the set S as the set of all permutations of the 9 letters given.
2. Let P_1, P_2, and P_3 be the properties that a permutation in S contains word MATH, IS, and FUN as consecutive letters, respectively.
3. For $i = 1, 2, 3$, let A_i be the set of those permutations in S satisfying property P_i.
4. $|S| = 9! = 362,880$.
5. The permutations in A_1 can be thought of as permutations of the six symbols, MATH, I, S, F, U, N. Hence $|A_1| = 6! = 720$.



The Inclusion-Exclusion Principle XIII

6. Similarly, the permutations in A_2 are permutations of the eight symbols, IS, M, A, T, H, F, U, N, and hence $|A_2| = 8! = 40,320$.
7. The permutations in A_3 are permutations of the seven symbols, FUN, M, A, T, H, I, S, and hence $|A_3| = 7! = 5,040$.
8. The permutations in $A_1 \cap A_2$ are permutations of the five symbols, MATH, IS, F, U, N. The permutations in $A_1 \cap A_3$ are permutations of the four symbols, MATH, FUN, I, S, and the permutations in $A_2 \cap A_3$ are permutations of the six symbols, IS, FUN, M, A, T, H. Hence

$$|A_1 \cap A_2| = 5! = 120,$$
$$|A_1 \cap A_3| = 4! = 24,$$
$$|A_2 \cap A_3| = 6! = 720.$$

9. Finally, $A_1 \cap A_2 \cap A_3$ consists of the permutations of the three symbols, MATH, IS, FUN. Hence $|A_1 \cap A_2 \cap A_3| = 3! = 6$.
10. Substituting into (6.1) we obtain

$$|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| = 362,880 - 720 - 40,320 - 5,040 + 120 + 24 + 720 - 6 = 317,658.$$



The Inclusion-Exclusion Principle XIV

A special case of the inclusion-exclusion principle

Assume that the size of the set $A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}$ that occurs in the inclusion-exclusion principle depends only on k and not on which k sets are used in the intersection. Thus there are constants $\alpha_0, \alpha_1, \alpha_2, \ldots, \alpha_n$ such that

$$\begin{aligned}
\alpha_0 &= |S|, \\
\alpha_1 &= |A_1| = |A_2| = \cdots = |A_m|, \\
\alpha_2 &= |A_1 \cap A_2| = \cdots = |A_{m-1} \cap A_m|, \\
\alpha_3 &= |A_1 \cap A_2 \cap A_3| = \cdots = |A_{m-2} \cap A_{m-1} \cap A_m|, \\
&\vdots \\
\alpha_m &= |A_1 \cap A_2 \cdots \cap A_m|.
\end{aligned}$$

In this case the inclusion-exclusion principle simplifies to

$$|\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_m}| = \alpha_0 - \binom{m}{1}\alpha_1 + \binom{m}{2}\alpha_2 - \binom{m}{3}\alpha_3 + \cdots$$



The Inclusion-Exclusion Principle XV

$$+(-1)^k\binom{m}{k}\alpha_k+\cdots+(-1)^m\alpha_m. \tag{6.3}$$

Example

How many integers between 0 and 99,999 (inclusive) have among their digits each of 2, 5, and 8?

1. Let S be the set of integers between 0 and 99,999. We think of the integers in S as the 5-permutations of the multiset in which each digit $0, 1, 2, \ldots, 9$ has repetition number 5 or greater.
2. Let P_1 be the property that an integer does not contain the digit 2, P_2 the property that an integer does not contain the digit 5, and P_3 the property that an integer does not contain the digit 8.
3. For $i = 1, 2, 3$, let A_i be the set consisting of those integers in S with property P_i.
4. We have

$$\alpha_0 = 10^5, \quad \alpha_1 = 9^5, \quad \alpha_2 = 8^5, \quad \alpha_3 = 7^5.$$



The Inclusion-Exclusion Principle XVI

5. By (6.3) the answer is

$$10^5 - 3 \times 9^5 + 3 \times 8^5 - 7^5.$$



Combinations with Repetition I

- The number of r-combinations of a set of n distinct elements is $\binom{n}{r}$.
- The number of r-combinations of a multiset with k distinct objects, each with an infinite repetition number, equals $\binom{r+k-1}{r}$.
- How this formula, in conjunction with the inclusion-exclusion principle, gives a method for finding the number of r-combinations of a multiset without any restrictions on its repetition numbers.
- We can summarize by saying that the number of r-combinations of a multiset $T = \{n_1 \cdot a_1, n_2 \cdot a_2, \cdots, n_k \cdot a_k\}$ can be determined in the two "extreme" cases:
  1. $n_1 = n_2 = \cdots = n_k = 1$; (i.e., T is a set) and
  2. $n_1 = n_2 = \cdots = n_k = r$.

Example

Determine the number of 10-combinations of the multiset $T = \{3 \cdot a, 4 \cdot b, 5 \cdot c\}$.

1. Apply the inclusion-exclusion principle to the set S of all 10-combinations of the multiset $T^* = \{\infty \cdot a, \infty \cdot b, \infty \cdot c\}$.



Combinations with Repetition II

2. Let P_1, P_2, and P_3 be the properties that 10-combinations of T^* have more than 3 a's, 4 b's, and 5 c's, respectively.
3. The number of 10-combinations of T is then the number of 10-combinations of T^* that have none of the properties P_1, P_2, and P_3.
4. As usual, let A_i consist of those 10-combinations of T^* which have property P_i $(i = 1, 2, 3)$.

$$|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| = |S| - (|A_1| + |A_2| + |A_3|) + (|A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3|) - |A_1 \cap A_2 \cap A_3|.$$

5. The size of S equals the number of 10-combinations of T^*.



Combinations with Repetition III

6. If we take any one of 10-combinations in A_1 and remove 4 a's, we are left with a 6-combination of T^*. Conversely, if we take a 6-combination of T^* and add 4 a's to it, we get a 10-combination of T^* in which a occurs at least 4 times. Thus the number of 10-combinations in A_1 equals the number of 6-combinations of T^*. And so on.

$$
\begin{aligned}
&|S| = \binom{10+3-1}{10} = 66,\\
&|A_1| = \binom{6+3-1}{6} = 28, \quad |A_2| = \binom{5+3-1}{5} = 21, \quad |A_3| = \binom{4+3-1}{4} = 15,\\
&|A_1 \cap A_2| = \binom{1+3-1}{1} = 3, \quad |A_1 \cap A_3| = \binom{0+3-1}{0} = 1, \quad |A_2 \cap A_3| = 0,\\
&|A_1 \cap A_2 \cap A_3| = 0.
\end{aligned}
$$

Putting all these results into the inclusion-exclusion principle, we obtain

$$
|\overline{A_1} \cap \overline{A_2} \cap \overline{A_3}| = 66 - (28 + 21 + 15) + (3 + 1 + 0) - 0 = 6
$$



Combinations with Repetition IV

7. The 6-combinations are

$$\{3 \cdot a, 4 \cdot b, 3 \cdot c\},$$
$$\{3 \cdot a, 3 \cdot b, 4 \cdot c\},$$
$$\{3 \cdot a, 2 \cdot b, 5 \cdot c\},$$
$$\{2 \cdot a, 4 \cdot b, 4 \cdot c\},$$
$$\{2 \cdot a, 3 \cdot b, 5 \cdot c\},$$
$$\{1 \cdot a, 4 \cdot b, 5 \cdot c\}.$$



Combinations with Repetition V

- We have already pointed out the connection between r-combinations and solutions of equations in integers. The number of r-combinations of the multiset $\{n_1 \cdot a_1, n_2 \cdot a_2, \ldots, n_k \cdot a_k\}$ equals the number of integral solutions of the equation

$$x_1 + x_2 + \cdots + x_k = r$$

which satisfy

$$0 \leq x_i \leq n_i \qquad (i = 1, 2, \ldots, k).$$



Combinations with Repetition VI

Example

What is the number of integral solution of the equation

$$x_1 + x_2 + x_3 + x_4 = 18$$

that satisfy

$$1 \leq x_1 \leq 5, -2 \leq x_2 \leq 4, 0 \leq x_3 \leq 5, 3 \leq x_4 \leq 9?$$



Combinations with Repetition VII

Solution

Introduce new variables,

$$y_1 = x_1 - 1, y_2 = x_2 + 2, y_3 = x_3, y_4 = x_4 - 3,$$

and our equation becomes

$$y_1 + y_2 + y_3 + y_4 = 16$$

with

$$0 \leq y_1 \leq 4, 0 \leq y_2 \leq 6, 0 \leq y_3 \leq 5, 0 \leq y_4 \leq 6.$$

The solution is obtained by applying the inclusion-exclusion principle

$$\begin{aligned}\binom{19}{16} - \left[\binom{14}{11} + \binom{12}{9} + \binom{13}{10} + \binom{12}{9}\right] \\ + \left[\binom{7}{4} + \binom{8}{5} + \binom{7}{4} + \binom{6}{3} + \binom{5}{2} + \binom{6}{3}\right] = 55.\end{aligned}$$



Derangements I

Definition

A *derangement* of $\{1, 2, \cdots, n\}$ ia a permutation $i_1 i_2 \cdots i_n$ of $\{1, 2, \cdots, n\}$ such that $i_1 \neq 1, i_2 \neq 2, \cdots, i_n \neq n$. We denote D_n the number of derangements of $\{1, 2, \cdots, n\}$.

Theorem

For $n \geq 1$,

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right).$$

1. Let S be the set of all $n!$ permutations of $\{1, 2, \ldots, n\}$.
2. For $j = 1, 2, \ldots, n$, let P_j be the property that, in a permutation, j is in its natural position.



Derangements II

3. Let A_j denote the set of permutations of $\{1, 2, \ldots, n\}$ with property P_j, $(j = 1, 2, \ldots, n)$.
4. The derangements of $\{1, 2, \ldots, n\}$ are precisely those permutations in $\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_n}$. Hence,

$$D_n = |\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_n}|,$$

and we use the inclusion-exclusion principle to evaluate D_n.

5. Generally,

$$|A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}| = (n-k)!$$

for any k-combination $\{i_1, i_2, \ldots, i_k\}$ of $\{1, 2, \ldots, n\}$.



Derangements III

6. Since there are $\binom{n}{k}$ k-combinations of $\{1, 2, \ldots, n\}$, we obtain

$$\begin{aligned} D_n = & n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! - \binom{n}{3}(n-3)! \\ & + \cdots + (-1)^n \binom{n}{n} 0! \\ = & n! - \frac{n!}{1!} + \frac{n!}{2!} - \frac{n!}{3!} + \cdots + (-1)^n \frac{n!}{n!} \\ = & n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right). \end{aligned}$$

Thus, the theorem is proved. □



Derangements IV

The derangement numbers D_n satisfy other relations that facilitate their evaluation.

$$D_n = (n-1)(D_{n-2} + D_{n-1}), \quad (n = 3, 4, 5, \cdots), \tag{6.4}$$

$$D_n = nD_{n-1} + (-1)^n \quad (n = 2, 3, 4, \cdots). \tag{6.5}$$

Formula (6.4) can be verified combinatorially.

1. Let $n \geq 3$, and consider the D_n derangements of $\{1, 2, \ldots, n\}$. These derangements can be partitioned into $n-1$ parts according to which of the integers $2, 3, \ldots, n$ is in the first position of the permutation.
2. It should be clear that each part contains the same number of derangements. Thus, D_n equals $(n-1)d_n$, where d_n is the number of derangements in which 2 is in the first position.
3. Such derangements are of the form

$$2i_2 i_3 \cdots i_n, \qquad i_2 \neq 2, i_3 \neq 3, \ldots, i_n \neq n.$$



Derangements V

4. These d_n derangements can be partitioned further into two subparts according as to whether $i_2 = 1$ or $i_2 \neq 1$. Let d'_n be the number of derangements of the form

$$21 i_3 \cdots i_n, \qquad i_3 \neq 3, \ldots, i_n \neq n.$$

Let d''_n be the number of derangements of the form

$$2 i_2 i_3 \cdots i_n, \qquad i_2 \neq 1, i_3 \neq 3, \ldots, i_n \neq n.$$

5. Thus $d_n = d'_n + d''_n$, and it follows that

$$D_n = (n-1)d_n = (n-1)(d'_n + d''_n).$$



Derangements VI

6. We observe that d'_n is the same as the number of permutations $i_3 i_4 \cdots i_n$ of $\{3, 4, \ldots, n\}$ in which $i_3 \neq 3, i_4 \neq 4, \ldots, i_n \neq n$. In other words, $d'_n = D_{n-2}$. Similarly, $d''_n = D_{n-1}$, and we conclude that

$$D_n = (n-1)(d'_n + d''_n) = (n-1)(D_{n-2} + D_{n-1}),$$

which is (6.4).

Example

At a party there are n men and n women. In how many ways can the n women choose male partners for the first dance? How many ways are there for the second dance if everyone has to change partners?

Solution

$n!$ and D_n.



Derangements VII

Example

Suppose the n men and the n women at the party check their hats before the dance. At the end of the party their hats are returned randomly. In how many ways can they be returned if each man gets a male hat and each woman gets a female hat, but no one gets the hats he or she checked?

Solution

$D_n \times D_n$.



Permutation with Forbidden Positions I

- Let

$$X_1, X_2, \ldots, X_n$$

be (possibly empty) subsets of $\{1, 2, \ldots, n\}$. We denote by

$$P(X_1, X_2, \ldots, X_n)$$

the set of all permutations $i_1 i_2 \cdots i_n$ of $\{1, 2, \ldots, n\}$ such that

$$i_1 \notin X_1,$$
$$i_2 \notin X_2, \qquad \vdots$$
$$i_n \notin X_n.$$

The number of permutations in $P(X_1, X_2, \cdots, X_n)$ is denoted by

$$p(X_1, X_2, \cdots, X_n) = |P(X_1, X_2, \cdots, X_n)|.$$



Permutation with Forbidden Positions II

Example

Let $n = 4$ and let $X_1 = \{1, 2\}$, $X_2 = \{2, 3\}$, $X_3 = \{3, 4\}$, $X_4 = \{1, 4\}$. Then $P(X_1, X_2, X_3, X_4)$ consists of all permutations $i_1 i_2 i_3 i_4$ of $\{1, 2, 3, 4\}$ such that

$$i_1 \neq 1, 2; \quad i_2 \neq 2, 3; \quad i_3 \neq 3, 4; \quad \text{and} \quad i_4 \neq 1, 4.$$

Solution

The set $P(X_1, X_2, X_3, X_4)$ contains only the two permutations 3412 and 4123. Thus, $p(X_1, X_2, X_3, X_4) = 2$.

Example

Let $X_1 = \{1\}$, $X_2 = \{2\}$, $\cdots$, $X_n = \{n\}$. Then the set $P(X_1, X_2, \cdots, X_4)$ equals the set of all permutations $i_1 i_2 \cdots i_n$ of $\{1, 2, \cdots, n\}$ for which $i_1 \neq 1, i_2 \neq 2, \cdots, i_n \neq n$. We conclude that $P(X_1, X_2, \cdots, X_4)$ is the set of derangements of $\{1, 2, \cdots, n\}$, and we have $p(X_1, X_2, \cdots, X_n) = D_n$.

Permutation with Forbidden Positions III

Example

Let $n = 5$ and let $X_1 = \{1, 4\}$, $X_2 = \{3\}$, $X_3 = \emptyset$, $X_4 = \{1, 5\}$, $X_5 = \{2, 5\}$. Then the permutations in $P(X_1, X_2, X_3, X_4, X_5)$ are in one-to-one correspondence with the placement of 5 nonattacking rooks on the board with forbidden positions as shown.

	1	2	3	4	5
1	×			×	
2			×		
3					
4	×				×
5		×			×

- For convenience, our argument will be couched in the language of nonattacking rooks on an n-by-n board.



Permutation with Forbidden Positions IV

- Let S be the set of all $n!$ placements of n nonattacking rooks on an n-by-n board. We say that such a placement of n nonattacking rooks satisfies property P_j provided that the rook in the jth row is in a column belonging to X_j $(j = 1, 2, \cdots, n)$. As usual, A_j denotes the set of rook placements satisfying property P_j, $(j = 1, 2, \cdots, n)$. The rooks that satisfy none of the properties $P_1, P_2, \cdots, P_n$. Hence,

$$\begin{aligned} p(X_1, X_2, \cdots, X_n) &= |\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_n}| \\ &= n! - \sum |A_i| + \sum |A_i \cap A_j| \\ &\quad - \cdots + (-1)^k \sum |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}| \\ &\quad + \cdots + (-1)^n |A_1 \cap A_2 \cap \cdots \cap A_n| \end{aligned} \tag{6.6}$$

  where the kth summation is over all k-combinations of $\{1, 2, \cdots, n\}$.
- What does, for instance, $|A_1|$ count?
  1. It counts the number of ways to place n nonattacking rooks on the board where the rook in row 1 is in one of the columns in X_1.



Permutation with Forbidden Positions V

2. We can choose the column of that rook in $|X_1|$ ways and then place the remaining $n-1$ nonattacking rooks in $(n-1)!$ ways.
3. Thus, $|A_1| = |X_1|(n-1)!$ and, more generally,

$$|A_i| = |X_i|(n-1)!, \qquad (i = 1, 2, \ldots, n).$$

4. Hence,

$$\sum |A_i| = (|X_1| + |X_2| + \cdots + |X_n|)(n-1)!.$$

5. Let $r_1 = |X_1| + |X_2| + \cdots + |X_n|$, we have

$$\sum |A_i| = r_1(n-1)!.$$

The number r_1 equals the number of forbidden squares of the board. Equivalently, r_1 equals the number of ways to place one rook on the board in a forbidden square.

- Now consider $|A_1 \cap A_2|$.



Permutation with Forbidden Positions VI

1. This number counts the number of ways to place n nonattacking rooks on the board where the rooks in row 1 and row 2 are both in forbidden positions (in X_1 and X_2, respectively).
2. Each placement of two nonattacking rooks in rows 1 and 2 in forbidden positions can be completed to n nonattacking rooks in $(n-2)!$ ways.
3. Similar considerations hold for any $|A_i \cap A_j|$, and, let r_2 equal the number of ways to place two nonattacking rooks on the board in forbidden positions. Then

$$\sum |A_i \cap A_j| = r_2(n-2)!.$$



Permutation with Forbidden Positions VII

- We may directly generalize the preceding argument and evaluate the kth sum in (6.6). We define r_k as follows,

  r_k is the number of ways to place k nonattacking rooks on the n-by-n board where each of the k rooks is in a forbidden position, $(k = 1, 2, \cdots, n)$.

  Then

$$\sum |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}| = r_k(n-k)!, \quad (k = 1, 2, \cdots, n).$$

Theorem

The number of ways to place n nonattacking, indistinguishable rooks on an n-by-n board with forbidden positions equals

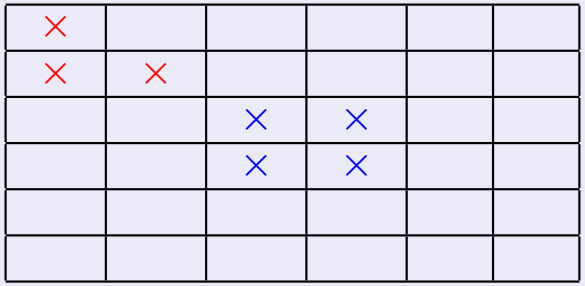
$$n! - r_1(n-1)! + r_2(n-2)! - \cdots + (-1)^k r_k(n-k)! + \cdots + (-1)^n r_n.$$



Permutation with Forbidden Positions VIII

Example

Determine the number of ways to place 6 nonattacking rooks on the following 6-by-6 board, with forbidden positions as shown.



Permutation with Forbidden Positions IX

Solution

We have

$$r_1 = 7;$$
$$r_2 = 1 + 2 + 3 \times 4 = 15;$$
$$r_3 = 1 \times 4 + 3 \times 2 = 10;$$
$$r_4 = 1 \times 2 = 2;$$
$$r_5 = r_6 = 0.$$

The number of ways to place six nonattacking rooks on the board so that no rook occupies a forbidden position equals

$$6! - 7 \times 5! + 15 \times 4! - 10 \times 3! + 2 \times 2! = 184.$$



Another Forbidden Position Problem I

- A problem of counting permutations in which there are certain *relative* forbidden positions.
- For each positive integer n, we let Q_n denote the number of permutations of $\{1, 2, \cdots, n\}$ in which none of the patterns $12, 23, \cdots, (n-1)n$ occurs. We use the inclusion-exclusion principle to evaluate Q_n.

$$Q_1 = 1, \quad Q_2 = 1, \quad Q_3 = 3, \quad Q_4 = 11.$$

n	Q_n	Permutations
1	1	1
2	1	2 1
3	3	2 1 3 3 2 1 1 3 2
4	11	4 1 3 2 4 3 2 1 4 2 1 3 3 2 1 4 3 2 4 1 2 1 4 3 2 4 3 1 2 4 1 3 3 1 4 2 1 3 2 4 1 4 3 2

Another Forbidden Position Problem II

Theorem

For $n \geq 1$,

$$Q_n = n! - \binom{n-1}{1}(n-1)! + \binom{n-1}{2}(n-2)! - \binom{n-1}{3}(n-3)! + \cdots + (-1)^{n-1}\binom{n-1}{n-1}1!.$$

1. Let S be the set of all $n!$ permutations of $\{1, 2, \ldots, n\}$.
2. Let P_j be the property that, in a permutation, the pattern $j(j+1)$ does occur, $(j = 1, 2, \ldots, n-1)$.
3. As usual, let A_j denote the set of permutations of $\{1, 2, \ldots, n\}$ that satisfy property P_j, $(j = 1, 2, \ldots, n-1)$. Then

$$Q_n = |\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_{n-1}}|,$$



Another Forbidden Position Problem III

4. A permutation is in A_1 if and only if the pattern 12 occurs in it. Thus, a permutation in A_1 may be regarded as a permutation of the $n-1$ symbols $\{12, 3, 4, \ldots, n\}$.
5. We conclude that $|A_1| = (n-1)!$, and in general

$$|A_j| = (n-1)! \qquad (j = 1, 2, \ldots, n-1).$$

6. Permutations that are in two of the sets $A_1, A_2, \ldots, A_{n-1}$ contain two patterns. These patterns either share an element, such as the patterns 12 and 23, or have no element in common, such as the patterns 12 and 34.
7. A permutation which contains the two patterns 12 and 34 can be regarded as a permutation of the $n-2$ symbols $\{12, 34, 5, \ldots, n\}$. Thus, $|A_1 \cap A_3| = (n-2)!$.
8. A permutation that contains the two patterns 12 and 23 contains the pattern 123 and thus can be regarded as a permutation of the $n-2$ symbols $\{123, 4, \ldots, n\}$. Hence, $|A_1 \cap A_2| = (n-2)!$.



Another Forbidden Position Problem IV

9. In general,

$$|A_i \cap A_j| = (n-2)!$$

for each 2-combination $\{i, j\}$ of $\{1, 2, \ldots, n\}$.

10. More generally,

$$|A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}| = (n-k)!$$

for each k-combination $\{i_1, i_2, \ldots, i_k\}$ of $\{1, 2, \ldots, n\}$.

11. Since, for each $k = 1, 2, \ldots, n-1$, there are $\binom{n-1}{k}$ k-combinations of $\{1, 2, \ldots, n-1\}$, applying the inclusion-exclusion principle we obtain the formula in the theorem. □

- Using the formula of the theorem, we calculate that

$$Q_5 = 5! - \binom{4}{1}4! + \binom{4}{2}3! - \binom{4}{3}2! + \binom{4}{4}1! = 53.$$



Another Forbidden Position Problem V

- The numbers $Q_1, Q_2, Q_3, \ldots$ are closely related to the derangement numbers. Indeed

$$Q_n = D_n + D_{n-1}, \qquad (n \geq 2).$$



Möbius Inversion I

- The inclusion-exclusion principle is an instance of Möbius inversion on a finite partially ordered set.

A more general version of the inclusion-exclusion principle

Let n be a positive integer and consider the set $X_n = \{1, 2, \cdots, n\}$ of n elements, and the partially ordered set $(\mathcal{P}(X_n), \subseteq)$ of all subsets of X_n partially ordered by containment. Let

$$F : \mathcal{P}(X_n) \to \mathcal{R}$$

be a real-valued function defined on $\mathcal{P}(X_n)$. We use F to define a new function

$$G : \mathcal{P}(X_n) \to \mathcal{R}$$

by

$$G(K) = \sum_{L \subseteq K} F(L), \qquad (K \subseteq X_n). \tag{6.7}$$

Möbius inversion allows one to *invert* equation (6.7) and to recover F from G; specifically, we have

$$F(K) = \sum_{L \subseteq K} (-1)^{|K|-|L|} G(L), \qquad (K \subseteq X_n). \tag{6.8}$$



Möbius Inversion II

- Let $A_1, A_2, \cdots, A_n$ be subsets of a finite set S, and for $K \subseteq X_n$, define $F(K)$ to be the number of elements of S that belong only to those set A_i with $i \notin K$. Thus, for $s \in S$, s is counted by $F(K)$ if and only if

$$s \notin A_i, \quad \text{for each } i \in K, \quad \text{and} \quad s \in A_j, \quad \text{for each } j \notin K.$$

Then

$$G(K) = \sum_{L \subseteq K} F(L)$$

counts the number of elements that belong to all of the sets A_j with j not in K and possibly other sets as well. Thus,

$$G(K) = |\cap_{i \notin K} A_i|.$$

By (6.8),

$$F(K) = \sum_{L \subseteq K} (-1)^{|K|-|L|} G(L). \tag{6.9}$$

Taking $K = X_n$ in (6.9), we get

$$F(X_n) = \sum_{L \subseteq X_n} (-1)^{n-|L|} G(L). \tag{6.10}$$

Möbius Inversion III

- Now, $F(X_n)$ counts the number of elements of S that belong only to those set A_i with $i \notin X_n$; that is, $F(X_n)$ is the number of elements of S that belong to none of sets $A_1, A_2, \cdots, A_n$ and thus equals the number of elements contained in $\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_n}$, Substituting into (6.10), we obtain

$$|\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_n}| = \sum_{L \subseteq X_n} (-1)^{n-|L|} |\cap_{i \notin L} A_i|,$$

or equivalently, by replacing L with its complement J in X_n,

$$|\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_n}| = \sum_{J \subseteq X_n} (-1)^{|J|} |\cap_{i \in J} A_i|. \tag{6.11}$$

Equation (6.11) is equivalent to the formula for the inclusion-exclusion principle.

